Multi-Phase Search Strategy and Results

Three-Phase Architecture I

This review employs a **three-phase search strategy** that progressively refines coverage from broad foundational literature through technology-specific studies to targeted molecular detection analyses. Each phase builds on prior phases through cross-phase citation validation and shared deduplication.

Three-Phase Architecture II

Phase 1: Exoplanet Atmosphere Foundation

Objective: Establish the foundational literature on exoplanet atmospheric studies, covering the broadest range of research from 2010 onwards.

Queries: 1. `"exoplanet atmosphere" OR "exoplanet atmospheric"` 2. `"transit spectroscopy" AND "atmosphere"` 3. `"atmospheric composition" AND "exoplanet"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2010 - Maximum year: 2026 - Minimum citation count: 0 (inclusive to capture recent work)

Results: 46 papers discovered, forming the foundational knowledge base for the review. This phase captures the broadest scope of atmospheric research, from observational characterizations to theoretical modeling studies.

Three-Phase Architecture III

Phase 2: James Webb Space Telescope Studies

Objective: Identify papers specifically utilizing JWST observational data for exoplanet atmospheric analysis, capturing the post-launch (2021+) surge in high-precision spectroscopic observations.

Queries: 1. `"James Webb" AND "atmosphere" AND ("exoplanet" OR "planet")` 2. `"JWST" AND "transit" AND "spectroscopy"` 3. `"NIRSpec" OR "MIRI" OR "NIRCam" AND "exoplanet"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2020 (capturing pre-launch calibration papers) - Maximum year: 2026

Dependencies: Builds on Phase 1 foundational papers.

Results: 17 papers after deterministic filtering. This phase targets records associated with JWST-era observations and analysis. Any claim about instrumental capability or scientific findings requires live, source-backed review.

Three-Phase Architecture IV

Phase 3: Molecular Detection Studies

Objective: Identify papers focused on detecting and analyzing specific atmospheric molecules (H O, CO , CH , H S, Na, K) in the configured domain.

Queries: 1. `"water vapor" AND "exoplanet" AND ("detection" OR "abundance")` 2. `"carbon dioxide" AND "exoplanet" AND ("CO2" OR "atmosphere")` 3. `"methane" AND "exoplanet" AND ("CH4" OR "atmosphere")` 4. `"hydrogen sulfide" OR "H2S" AND "exoplanet"` 5. `"sodium" OR "potassium" AND "exoplanet atmosphere"`

Engines: arXiv, OpenAlex, Crossref, Semantic Scholar

Deterministic Filters: - Minimum year: 2015 - Maximum year: 2026

Dependencies: Builds on both Phase 1 and Phase 2 papers.

Results: 29 papers covering molecular detection across diverse exoplanet populations, from hot Jupiters to terrestrial worlds.

Cross-Phase Analysis I

Phase Overlap

The three phases produced complementary but overlapping corpora, with 49.5% average Jaccard similarity between phases. Papers discovered in multiple phases are tracked with full provenance, enabling analysis of papers that span multiple research paradigms.

Citation Validation

Cross-phase citation analysis records that 39.4% of later-phase papers cite foundational work from Phase 1. This is a descriptive connection statistic for the retained corpus; it does not establish methodological coherence across the field.

Combined Corpus

The deduplicated combined corpus contains 46 unique papers spanning 2010–2023 (13 years). It provides a multi-phase description of the retained retrieval slice, not a field-wide account.

LLM-Based Content Filtering I

Three LLM-based content filters were designed for this review:

1. **Study Type Classification**: Categorizes papers as observational, theoretical, review, or other. Keeps only observational and theoretical studies for analysis.
2. **JWST Data Analysis**: Identifies papers that analyze actual JWST observational data, distinguishing from theoretical JWST predictions.
3. **Molecular Detection Focus**: Filters for papers primarily focused on detecting and measuring specific atmospheric molecules.

These filters are configurable through `manuscript/config.yaml` and can be enabled for specific phases or applied across the entire corpus. When enabled, they use a local Ollama LLM instance for cost-effective, privacy-preserving content classification.